BRAIN AS A CPU SYSTEM

1. Brain as a CPU System

- The human brain can be thought of as a **highly sophisticated and complex** information processing system, similar to a computer's Central Processing Unit (CPU).
- Both the brain and CPU **receive and process inputs**, **store information**, and **perform calculations to produce outputs.**

- Just like how a computer's CPU has an arithmetic logic unit (ALU) to perform mathematical calculations, the human brain has specialized regions for processing mathematical and logical operations.
- The **prefrontal cortex**, for example, is responsible for higher-level cognitive functions such as **decision making and problem solving.**

- Similarly, a computer's CPU also has **memory units** for storing information, and the human brain has several regions dedicated to memory storage, including the **hippocampus and amygdala**

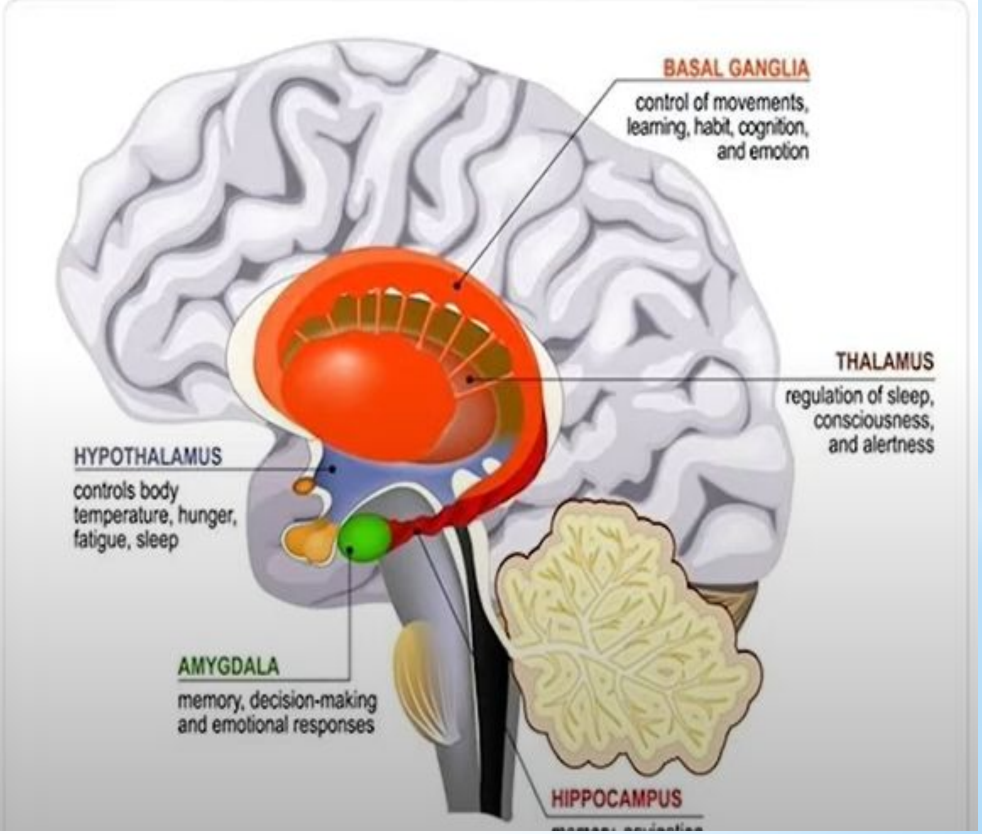
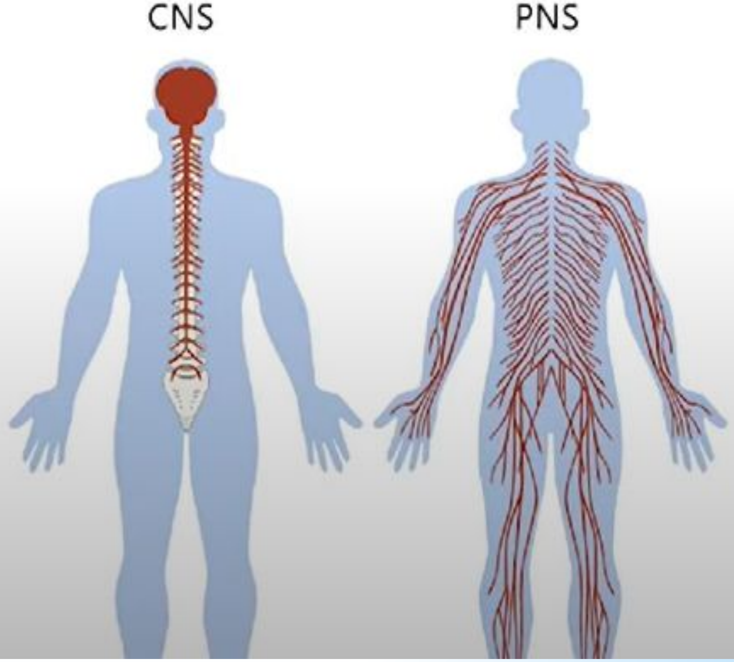


Figure 2.3: Limbic system. Cross section of the human brain. Mammillary body, basal ganglia, pituitary gland, amygdala, hippocampus, thalamus - Illustration Credit: Designua / Shutterstock

1.2 CNS and PNS

- The Central Nervous System (CNS) and Peripheral Nervous System (PNS) are the two main components of the nervous system in the human body.



CNS

- The Central Nervous System consists of the brain and spinal cord and is responsible for receiving, processing, and integrating sensory information and transmitting commands to the rest of the body.
- The brain acts as the command center, receiving and processing sensory inputs and generating motor outputs, while the spinal cord acts as a relay center, transmitting information between the brain and peripheral nerves.

PNS

- The Peripheral Nervous System, on the other hand, consists of all the nerves that lie outside the brain and spinal cord.
- It is responsible for transmitting sensory information from the periphery of the body (such as the skin, muscles, and organs) to the CNS, and transmitting commands from the CNS to the periphery.
- The PNS can be further divided into the somatic nervous system and the autonomic nervous system.

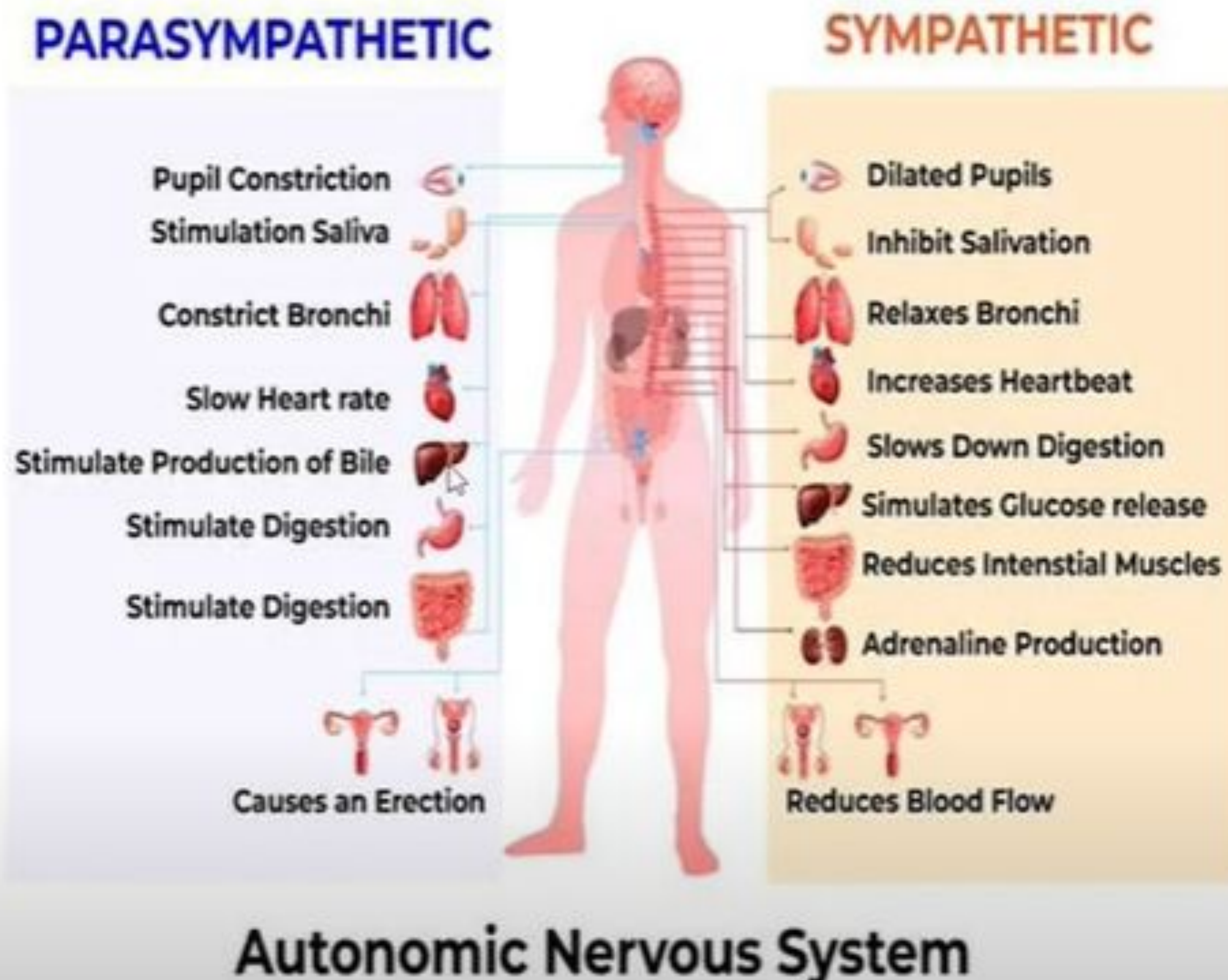
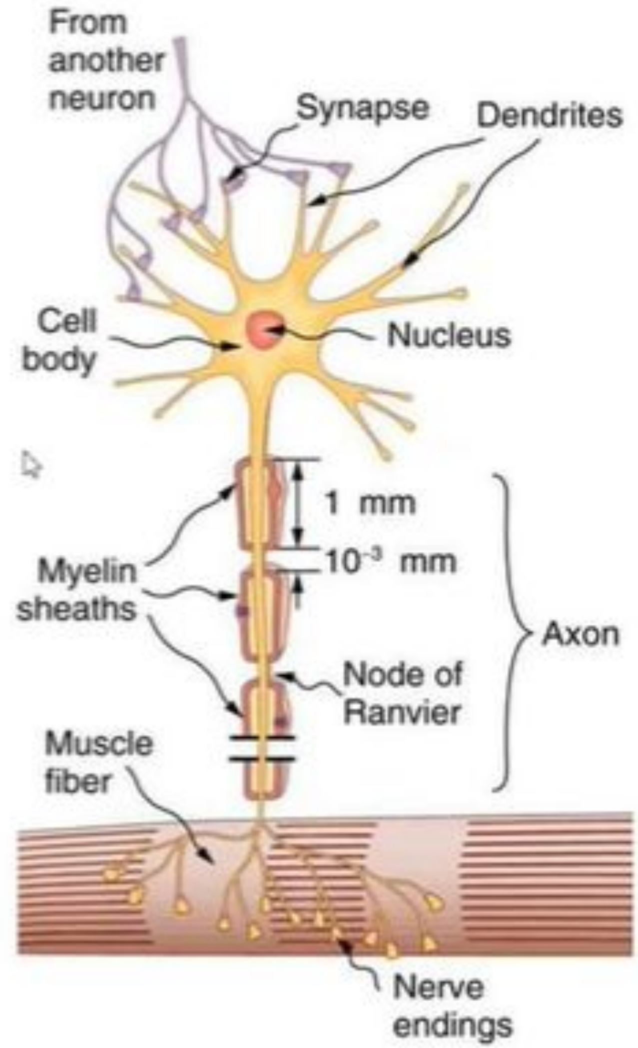


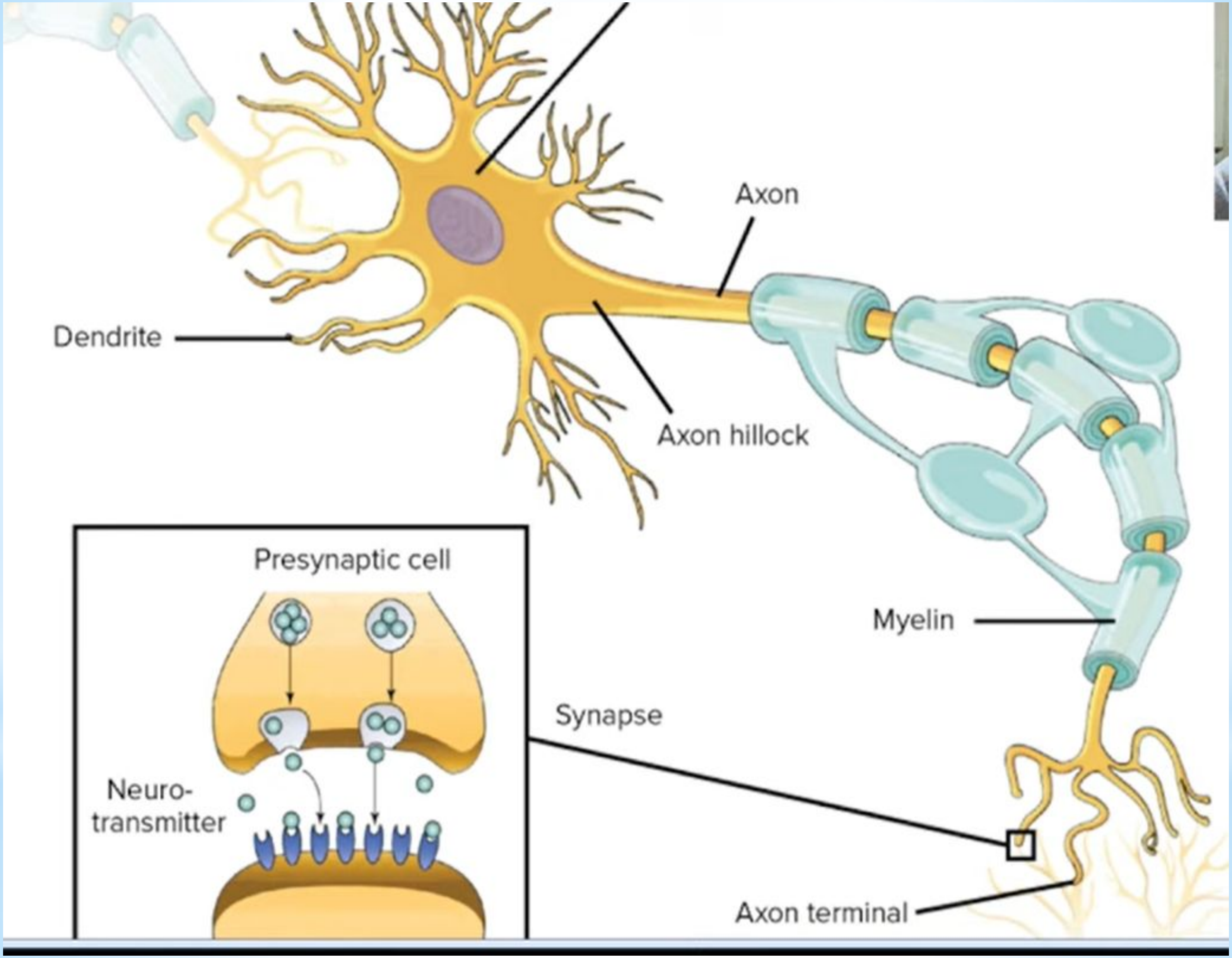
Figure 2.5: Representation of function of autonomic nervous system

SIGNAL TRANSMISSION IN BRAIN

SIGNAL TRANSMISSION IN BRAIN

A neuron or nerve cell is the fundamental unit of the nervous system. Neurons are responsible for receiving, processing, and transmitting electrical and chemical signals within the nervous system. They communicate and coordinate between different parts of the body and mediate various physiological and cognitive processes.





Axon
Dendrite
Axon hillock
Myelin
Presynaptic cell
Synapse
Neuro-
transmitter
Axon terminal

Figure: Representing the process of transmission of information through nerve cells (syn
transmission)

A neuron receives inputs from other neurons at its dendrites, integrates the information and then generates an electrical impulse, or action potential, that travels down its axon to the synaptic terminals. At the synaptic terminals, the neuron releases chemical neurotransmitters, which cross the synaptic gap and bind to receptors on the postsynaptic neuron, leading to the initiation of another action potential in the postsynaptic neuron.

This process of transmitting information from one neuron to another is known as synaptic transmission and forms the basis of communication within the brain.

Different types of neurotransmitters have different effects on postsynaptic neurons, and the balance of neurotransmitter levels can influence brain function, including mood, learning, and memory.

Signal transmission in the brain is also influenced by various forms of synaptic plasticity, including long-term potentiation (LTP) and long-term depression (LTD), which can modify the strength of synaptic connections and contribute to learning and memory processes.

Basis for Comparison	Brain	Computer
Construction	Neurons and synapses	ICs, transistors, diodes, capacitors, transistors, etc.
Memory growth	Increases each time by connecting synaptic links	Increases by adding more memory chips
Backup systems	Built-in backup system	Backup system is constructedmanually
Memory power	100 teraflops (100 trillion calculations/seconds)	100 million megabytes
Memory density	10^7 circuits/cm^3	10^{14} bits/cm^3
Energy consumption	12 watts of power	Gigawatts of power
Information storage	Stored in electrochemical and electric impulses.	Stored in numeric and symbolic form (i.e. in binary bits).
Size and weight	The brain's volume is 1500 cm^3 and weight is around 3.3pounds.	Variable weight and size form few grams to tons.
Transmission of information	Uses chemicals to fire the action potential in the neurons.	Communicationis achieved through electrical coded signals.
Information processingpower	Low	High
Input/output equipment	Sensory organs	Keyboards, mouse, web cameras, etc.
Structural organization	Self-organized	Pre-programmed structure
Parallelism	Massive	Limited
Reliability and damageability properties	Brain is self-organizing, self-maintaining and reliable.	Computers perform a monotonous job and can't correct itself.

EEG stands for electroencephalography, which is a non-invasive method for measuring the electrical activity of the brain. An EEG records the electrical signals generated by the brain neurons as they communicate with each other. The signals are recorded through electrodes placed on the scalp and the resulting EEG pattern provides information about the synchronised electrical activity of large populations of neurons.

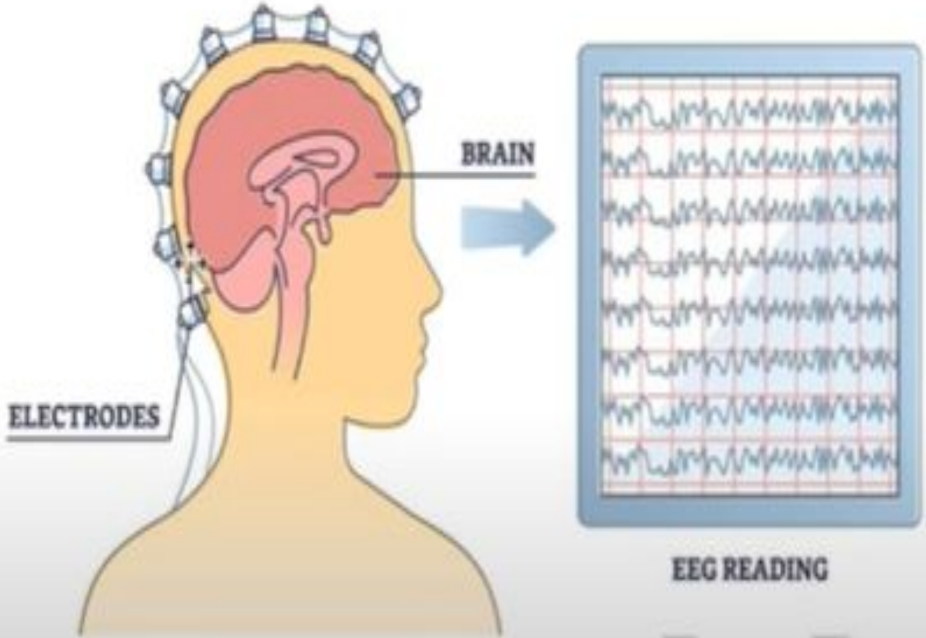


Figure: Representing EEG

Applications of EEG

- Diagnosis of Epilepsy: EEG is a widely used tool to diagnose epilepsy and other brain disorders. It can detect abnormal electrical activity in the brain, which can help to the diagnosis and determine the location of the seizure focus.
- Sleep Studies: EEG is often used in sleep studies to evaluate sleep patterns and diagnosis sleep disorders.
- Brain-Computer Interfaces (BCI): EEG can be used to control external devices such as prosthetic limbs or computer software. This is done by detecting specific brain waves associated with a particular mental state, such as concentration or relaxation.
- Research on Brain Function: EEG is used in research to study brain function during various activities such as reading, problem-solving, and decision-making. EEG can also be used to investigate how the brain responds to stimuli such as light, sound, and touch.
- Diagnosis of Brain Disorders: EEG can be used to diagnose a wide range of brain disorders including dementia, Parkinson's disease, and traumatic brain injury.
- Anesthesia Monitoring: EEG can be used to monitor the depth of anesthesia during surgery to ensure that the patient remains in a safe and comfortable state.
- Monitoring Brain Activity during Coma: EEG is also used to monitor brain activity in patients who are in a coma to determine the level of brain function and assess the likelihood of recovery.

Delta	1-4	Deep sleep
Theta	4-8	Drowsy
Alpha	8-12	Relaxed
Beta	12-30	Focused

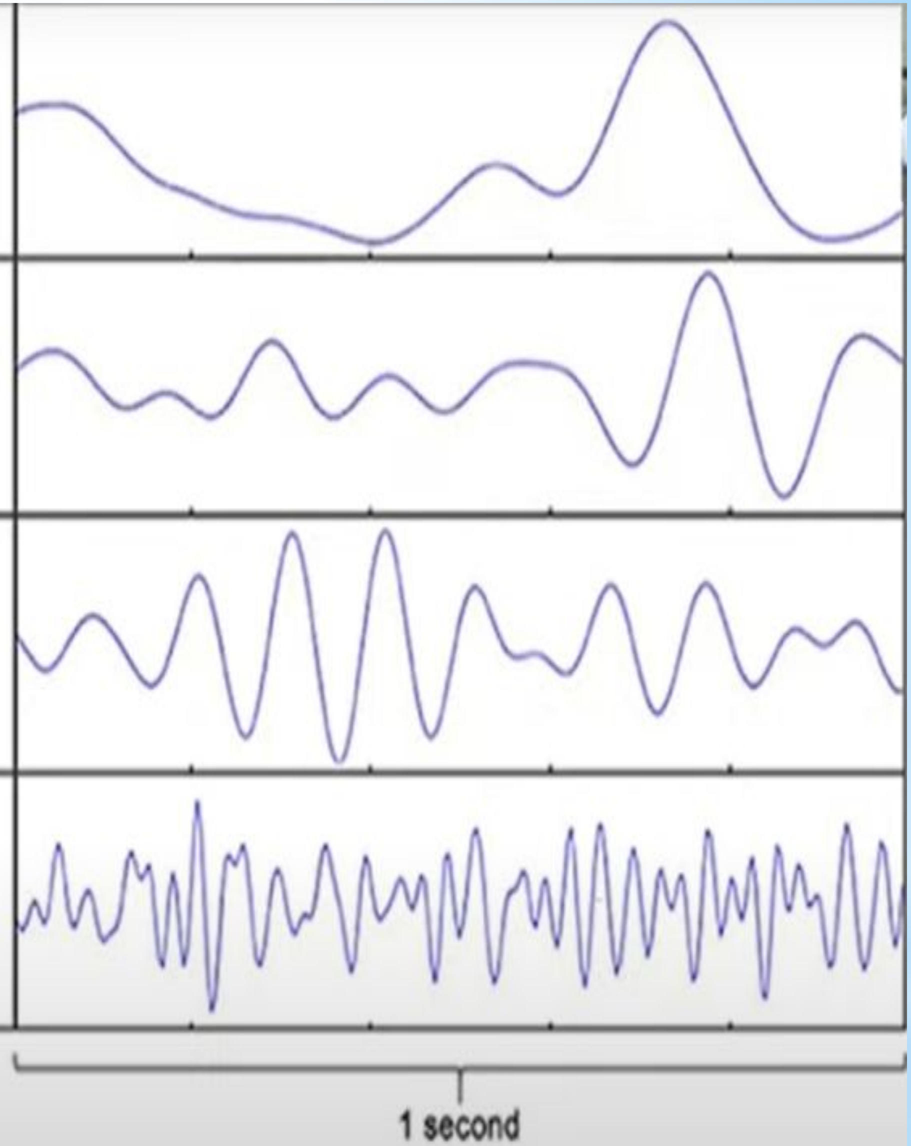


Figure: Representing EEG signal and the mental state of brain

- Delta waves (0.5-4 Hz): Delta waves are low-frequency waves associated with deep
  infancy, and brain disorders such as brain damage or dementia.
- Theta waves (4-8 Hz): Theta waves are also associated with sleep and relaxation, as went as meditation and hypnosis. They are also present during memory encoding and retrieval processes.
- Alpha waves (8-12 Hz): Alpha waves are present when the brain is relaxed and not focused on any particular task. They are also associated with meditation and creativity.
- Beta waves (12-30 Hz): Beta waves are present when the brain is focused on a task, such as problem-solving or decision-making. They are also associated with anxiety and stress.
- Gamma waves (30-100 Hz): Gamma waves are associated with high-level cognitive processing, such as attention, perception, and memory. They are also involved in sensory processing and motor control.

The analysis of EEG signals can provide valuable information about brain function and activity, as well as offer insights into the workings of the human mind.

Frequency band	Speed (Hz)	Mental state	Electroencephalography (EEG) recording

2.1.5 Robotic Arms for Prosthetics

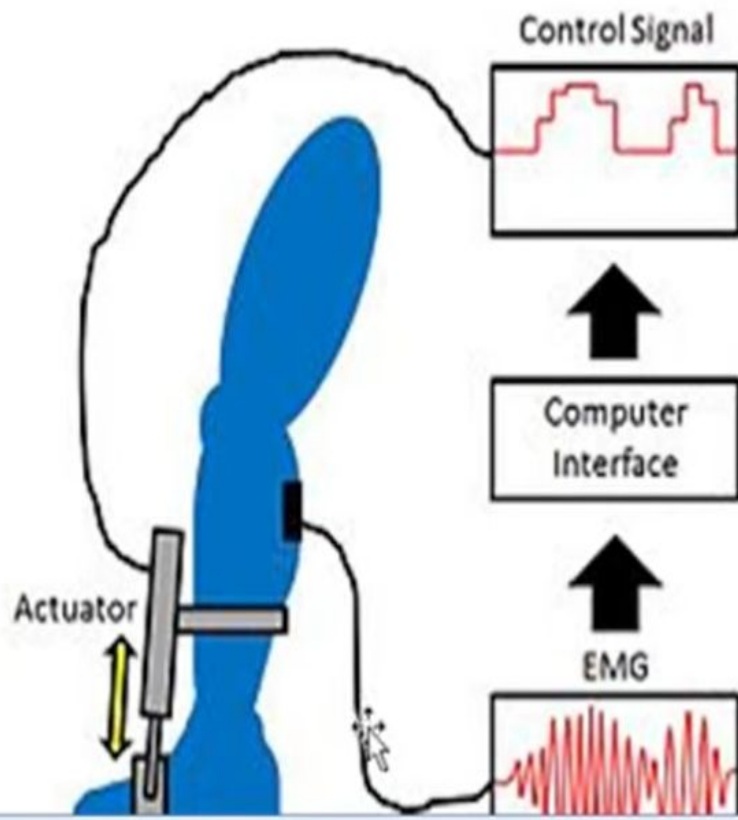
Robotic arms for prosthetics are advanced prosthetic devices that use robotics technics to restore functionality to individuals with upper limb amputations.

These devices typically use motors, actuators, and sensors to mimic the movements of a human arm and hand, allowing the wearer to perform tasks such as reaching, grasping, and manipulating objects.

Robotic arms for prosthetics can be controlled in a variety of ways, including direct control through muscle signals (myoelectric control) or brain-machine interfaces, which use electrodes implanted in the brain or placed on the scalp to detect and interpret brain activity.

Robotic Arm Prosthetic Direct Control through Muscle Signals (myoelectric control)

Myoelectric control of a robotic arm prosthetic involves using the electrical signals generated by the wearer's remaining muscles to control the movement of the prosthetic system typically involves electrodes placed on the skin over the remaining muscle that are used to detect and interpret the electrical signals generated by the muscle contractions.



2.1.6 Engineering Solutions for Parkinson's Disease

Parkinson's disease is a neurodegenerative disorder that affects movement and **motor** function. There are several engineering solutions aimed at improving the quality of **life** individuals with Parkinson's disease, including:

- Deep Brain Stimulation (DBS): DBS involves the implantation of electrodes into specific regions of the brain to deliver electrical stimulation, which can help to relieve symptoms such as tremors, stiffness, and difficulty with movement.
- Exoskeletons: Exoskeletons are wearable devices that provide support and assistance for individuals with mobility issues. Some exoskeletons have been developed specifically for people with Parkinson's disease, and can help to improve balance, reduce tremors, and increase overall mobility.
- Telerehabilitation: Telerehabilitation involves the use of telecommunication technology to provide physical therapy and rehabilitation services to individuals with Parkinson's disease, without the need for in-person visits to a therapist.
- Smartwatch Applications: Smartwatch applications can be used to monitor symptoms of Parkinson's disease, such as tremors, and provide reminders and prompts for medication and exercise.

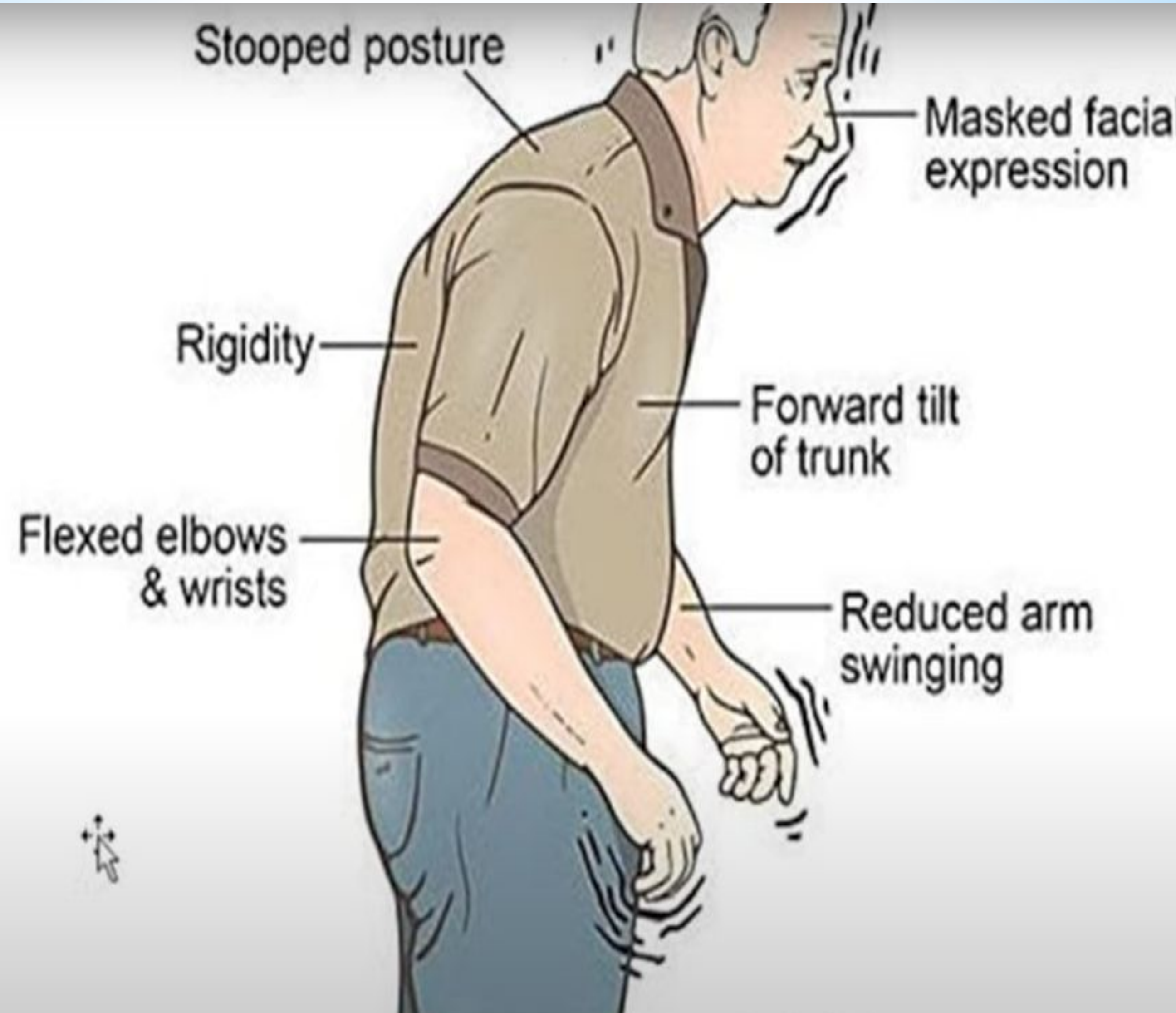


Figure: Representing typical appearance of Parkinson's disease

- Virtual Reality: Virtual reality systems can be used for rehabilitation and therapy for individuals with Parkinson's disease, providing interactive and engaging environments for patients to practice movements and improve coordination and balance.

These engineering solutions have the potential to significantly improve the quality of life for individuals with Parkinson's disease, and ongoing research and development is aimed at improving their effectiveness and accessibility. However, it is important to note that these technologies are not a cure for Parkinson's disease and should be used in conjunction with other forms of treatment and care.